

SMe Note

Acknowledge: Sample Space: Ω Random Variables: $\mathbf{Y} = (Y_1, \dots, Y_n)$ Sample Data: $\mathbf{y} = (y_1, \dots, y_n)$ $f_{\mathbf{Y}}(\mathbf{y}; \theta) = \prod_i f_{Y_i}(y_i; \theta)$, if Y_i are independent Distribution Function: $F_{\mathbf{Y}}(\mathbf{y}) = \mathbb{P}(Y_1 \leq y_1, \dots, Y_n \leq y_n)$
Parameter Space: Θ CDF: $F_{\mathbf{Y}}(\mathbf{y})$ PDF/PMF: $f_{\mathbf{Y}}(\mathbf{y})$ or $p_{\mathbf{Y}}(\mathbf{y})$

1 Basic Knowledge

Def of expectation (Moments): For any well-behaved function h on $\mathbb{R}$. $\mathbb{E}[h(Y)] := \sum_i h(y_i) \mathbb{P}(Y = y_i)$ $\mathbb{E}[h(Y)] := \int_{\mathbb{R}} h(y) f_Y(y) dy$

Def of Model: A collection of distributions indexed by a parameter $\{F_{\mathbf{Y}}(\cdot|\theta); \theta \in \Theta\}$ is model.

Likelihood Function: $L(\theta; \mathbf{y}) = f_{\mathbf{Y}}(\mathbf{y}; \theta)$ **MLE:** $\hat{\theta}(\mathbf{y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{y})] = [U(\hat{\theta}) = 0]$ ps: MLE has invariant property

(Strong) Likelihood Principle: $L(\theta; x) \propto L(\theta; y)$ [Strong: $L_f(\theta; x) \propto L_g(\theta; z)$; f, g are statistical model] $\Rightarrow$ Same Inferential Conclusions.

Def of Identifiability: A model is identifiable if $f(y; \theta_1) = f(y; \theta_2), \forall y$ then $\theta_1 = \theta_2$ (i.e. $\text{map}: \theta \rightarrow f(y; \theta)$ is 1-1)

• Method: 1.By def $\Rightarrow$ ratio not depend on $y \Rightarrow \theta_1, \theta_2$ relationship. (with any additional constraint) 2.MLE not unique $\Rightarrow$ not identifiable 3.Give an example θ 4.Fisher information

2 Properties of the Likelihood and the MLE

Positive [Semi-]definite matrix M: $\forall \mathbf{z} \in \mathbb{R}^n \setminus \mathbf{0}, \mathbf{z}^T \mathbf{M} \mathbf{z} > 0 [\geq 0] \Leftrightarrow (> 0)^1 n \text{ positive eigenvalues. } ^2 \text{ full rank. } ^3 \text{ invertible}$

Score Func / Vector U	Single: $U(\theta, \mathbf{Y}) = \ell'(\theta; \mathbf{Y})$	$\mathbb{E}[U] = \mathbf{0}$	$Var[U] = I(\theta)$
	Multiple: $\mathbf{U}(\theta) = [\partial_{\theta_1} \ell(\theta; \mathbf{Y}), \dots, \partial_{\theta_p} \ell(\theta; \mathbf{Y})]^T \in \mathbb{R}^p$	$\mathbb{E}[\mathbf{U}] = \mathbf{0}$	$Var[\mathbf{U}] = I(\theta)$ $I_{j,k} = -Cov(\frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_j}, \frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_k}) = -\mathbb{E}[\frac{\partial^2 \ell(\theta, \mathbf{Y})}{\partial \theta_j \partial \theta_k}]$
CLT of U	Single: $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, I(\theta_0)), n \rightarrow \infty$		
	Multiple: $\mathbf{U}(\theta_0; \mathbf{Y}) \sim \mathcal{N}_p(\mathbf{0}, I(\theta_0)), n \rightarrow \infty$	Also: $\mathbf{U}(\theta_0; \mathbf{Y})^T I^{-1}(\theta_0) \mathbf{U}(\theta_0; \mathbf{Y}) \sim \chi_p^2$	
Fisher Information I	Single: Observed: $J(\theta, \mathbf{y}) = -U'(\theta, \mathbf{y}) = -\ell''(\theta; \mathbf{y})$	Expected: $I(\theta) = -\mathbb{E}[\ell''(\theta; \mathbf{Y})]$ Unit: $i(\theta) = I(\theta)/n = -\mathbb{E}[\ell''(\theta; Y_i)]$	
	Multiple: Observed: $\mathbf{J}(\theta; \mathbf{y}) \in \mathbb{R}^{p \times p}, J_{j,k} = -\frac{\partial^2 \ell(\theta, \mathbf{y})}{\partial \theta_j \partial \theta_k}$	Expected: $I(\theta) \in \mathbb{R}^{p \times p}, I_{j,k} = -\mathbb{E}[\frac{\partial^2 \ell(\theta, \mathbf{Y})}{\partial \theta_j \partial \theta_k}]$ ps: $I(\theta)$ always positive semi-definite	
$I \uparrow \Rightarrow \mathbf{Y}$ 对 θ 越敏感; 信息量大; 参数估计更精确		A model is <i>identifiable</i> iff $I(\theta)$ is positive definite. ps:(一维) identifiable if $I(\theta) \neq 0$	
CLT of MLE CLT of I	Single: $I(\theta_0) < \infty \Rightarrow \hat{\theta} \sim \mathcal{N}(\theta_0, I^{-1}(\theta_0)), n \rightarrow \infty$	$\widehat{Var}(\hat{\theta}) \approx I^{-1}(\hat{\theta})$	$ESE(\hat{\theta}) \approx \sqrt{I^{-1}(\hat{\theta})}$ estimated standard error
	Multiple: $I(\theta_0) < \infty \Rightarrow \hat{\theta} \sim N_p(\theta_0, I^{-1}(\theta_0)), n \rightarrow \infty$	Also: $(\hat{\theta} - \theta_0)^T I(\theta_0) (\hat{\theta} - \theta_0) \sim \chi_p^2$	

2.1 For Single Parameter

*Random Variable / Function: $L(\theta; \mathbf{Y})$ is random function (NOT pdf). For fixed θ , it's r.v. $\hat{\theta}(\mathbf{Y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{Y})]$ is r.v. ps: I^{-1}, J^{-1} 是指其负一次方, 不是反函数 True Parameter: θ_0

Name	Test Statistic	Hypothesis	CI	Other
Score Test	Given $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, I(\theta_0))$ $T_{Score}(\mathbf{Y}) = \frac{U(\theta_0)}{\sqrt{I(\theta_0)}} \sim \mathcal{N}(0, 1)$	$H_0: \theta = \theta_0$ 不用 MLE, 泰勒展开	$\left\{ \theta: \frac{U^2(\theta)}{I(\theta)} < z_{\alpha, 1}^2 \right\}$	or $T_{Score}^2(\mathbf{Y}) \sim \chi_1^2$ $T_{Score}(\mathbf{Y}) \approx \frac{U(\theta_0)}{\sqrt{I(\hat{\theta})}}, \frac{U(\theta_0)}{\sqrt{J(\theta_0)}}, \frac{U(\theta_0)}{\sqrt{J(\hat{\theta})}}$
Wald Test or Maximum Likelihood(ML) Test	$T_{Wald}(\mathbf{Y}) = \sqrt{I(\theta_0)}(\hat{\theta} - \theta_0) = \frac{\hat{\theta} - \theta_0}{\sqrt{I^{-1}(\theta_0)}}$ $T_{Wald}(\mathbf{Y}) \sim \mathcal{N}(0, 1)$	$H_0: \theta = \theta_0$ 构建 CI	$\{ \theta: (\hat{\theta} - \theta)^2 I(\theta) < z_{\alpha, 1}^2 \}$	or $T_{Wald}^2 \sim \chi_1^2; n \rightarrow \infty T_{Wald} \simeq T_{Score}$ $T_{Wald}(\mathbf{Y}) \approx \frac{(\hat{\theta} - \theta_0)}{\sqrt{I^{-1}(\hat{\theta})}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{J^{-1}(\theta_0)}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{J^{-1}(\hat{\theta})}}$
log-Likelihood Ratio Test	$T_{LR}(\mathbf{Y}) = 2(\ell(\hat{\theta}) - \ell(\theta_0)) \sim \chi_1^2$ as $n \rightarrow \infty$ $LR = L(\theta_0)/L(\hat{\theta}), T_{LR} = -2\ln(LR)$	$H_0: \theta = \theta_0$ 最小 CI, Most Powerful	$\{ \theta: 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha, 1}^2 \}$	ps: $2(\ell(\hat{\theta}) - \ell(\theta_0)) \simeq (\theta_0 - \hat{\theta})^2 J(\hat{\theta})$
$T_{Score} \simeq T_{Wald} \simeq T_{LR}$ as $n \rightarrow \infty$; $J(\theta_0) \approx I(\theta_0)$ as $n \rightarrow \infty$; When $\ell(\theta)$ 是关于 θ 的二次函数 $\Rightarrow$ All same. Geometric: (T_S , 比斜率), (T_W , 比宽度), (T_{LR} , 比高度) ps: $z_{\alpha, n} = \chi_{\alpha, n}^2$				

2.2 For Single Parameter (Numerical Method)

Suppose initial value solution θ_0	Newton-Raphson Method ps: also called Newton's method.	Fisher's Method of Scoring ps: also called scoring method
Formula	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\ell''(\theta_r)} = \theta_r + \frac{U(\theta_r)}{J(\theta_r)}$	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\mathbb{E}[\ell''(\theta_r)]} = \theta_r + \frac{U(\theta_r)}{I(\theta_r)}$
Advantages	Converge <i>very quickly</i> near optimum	I is constant / easier to calculate; Converge for wider θ_0
Stopping Rule	Given tolerance level ε : ¹ Stop after r if $ U(\theta_r) \leq \varepsilon$ (Common) or ² Stop after $ \theta_r - \theta_{r-1} \leq \varepsilon$	

2.3 For Multiple Parameters

*此节中, θ 代表向量, $\theta = (\theta_1, \dots, \theta_p)$ I^{-1} 指其逆矩阵 True Parameter: θ_0

Name	Test Statistic ($T_{Score}^2, T_{Wald}^2, T_{LR}$)	Hypothesis	(Confidence Region) CR	Marginal Normal distribution $\hat{\theta}$: $\hat{\theta}_j \sim \mathcal{N}(\theta_j, [I^{-1}(\underline{\theta}_0)]_{jj})$ θ_j : true para • $ESE(\hat{\theta}_j) = \sqrt{[I^{-1}(\hat{\theta})]_{jj}}$ • $CI(\theta_j): \hat{\theta}_j \pm z_{\alpha} ESE(\hat{\theta}_j)$
Score Test	$\mathbf{U}(\theta_0)^T I^{-1}(\theta_0) \mathbf{U}(\theta_0) > z_{\alpha, p}$	$\theta = \theta_0$	$\{ \theta: \mathbf{U}(\hat{\theta})^T I^{-1}(\theta) \mathbf{U}(\hat{\theta}) < z_{\alpha, p} \}$	
Wald Test	$(\hat{\theta} - \theta_0)^T I(\theta_0) (\hat{\theta} - \theta_0) > z_{\alpha, p}$	$\theta = \theta_0$	$\{ \theta: (\hat{\theta} - \theta)^T I(\theta) (\hat{\theta} - \theta) < z_{\alpha, p} \}$	
log-Likelihood Ratio Test	$2(\ell(\hat{\theta}) - \ell(\theta_0)) > z_{\alpha, p}$	$\theta = \theta_0$	$\{ \theta: 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha, p} \}$	
$I(\theta_0) \approx I(\hat{\theta}), J(\theta_0), J(\hat{\theta})$ $ps: z_{\alpha, p} = \chi_{\alpha, p}^2$ T_{LR} : simple hypothesis testing. ($\dim(\mathcal{A}) = 0$) θ 都是向量				

Composite Hypothesis Test: $H_0: \underline{\theta} \in \mathcal{A}, H_1: \underline{\theta} \notin \mathcal{A}$ $\mathcal{A}$: reduced model $LR = \frac{\max_{\theta \in \mathcal{A}} L(\theta)}{\max_{\theta \in \Theta} L(\theta)} = \frac{L(\hat{\theta}_{\mathcal{A}})}{L(\hat{\theta}_{\Theta})}$ ps: 多维未知参数, H_0 只想其中几个时用 || $\dim(\mathcal{A})$ 常数时为 0

• **Generalised Likelihood Ratio Test:** $T_{GLR} = -2\ln(LR) = 2(\ell(\hat{\theta}) - \ell(\hat{\theta}_{\mathcal{A}})) \sim \chi_{p-r}^2$ $p: \#$ of parameters. $r = \dim(\mathcal{A})$

2.4 For Multiple Parameter (Numerical Method)

Method (start at θ_0)	Newton-Raphson Method	Fisher's Method of Scoring
Formula	$\theta_{r+1} = \theta_r - \mathbf{H}^{-1}(\theta_r) \mathbf{U}(\theta_r) = \theta_r + \mathbf{J}^{-1}(\theta_r) \mathbf{U}(\theta_r)$	$\theta_{r+1} = \theta_r - (\mathbb{E}[\mathbf{H}(\theta_r)])^{-1} \mathbf{U}(\theta_r) = \theta_r + I^{-1}(\theta_r) \mathbf{U}(\theta_r)$
ps: Jacobian / Hessian $\mathbf{H} = -\mathbf{J}$ If $I = \text{diag}(a, b, \dots) \Rightarrow I^{-1} = \text{diag}(a^{-1}, b^{-1}, \dots)$		

3 Bayesian Inference

3.1 Bayesian Approach & Definitions

Basic Idea: Consider θ as a r.v.. Bayesian approach $\Rightarrow$ Probability are used to represent degree of believe about the unknown θ

Prior Distribution $p(\theta)$: Prior distribution $p(\theta)$ contains all available prior information about θ before observing any data.

· **Hyper-Parameters:** 由于 $p(\theta)$ 是已知, 服从某种分布 (with parameter γ). We denoted $p(\theta)$ as $p(\theta; \gamma)$ where γ is Hyper-parameters. 定义: $\gamma_{prior} \rightarrow \gamma_{post}$

Likelihood function $p(y|\theta) = f(y|\theta)$: Any observed data is considered as a realisation of r.v. Y with $f(y|\theta)$

Posterior Distribution $p(\theta|y)$: Posterior distribution $p(\theta|y)$, contains all available knowledge.

· **Bayes' Rule / Theorem:** $p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{f(y|\theta)p(\theta)}{m(y)}$ $p(y) = m(y) = \int_{\theta \in \Theta} f(y|\theta)p(\theta)d\theta$ (Normalizing constant)

Name	MAPMaximum A Posteriori	MMSEPosterior Mean / Bayesian Minimum Mean Squared Error Estimator	Posterior Median $\hat{\theta}_{median}$
Single Para	$\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta y)$	$\hat{\theta}_{MMSE} = E[\theta y] = \int_{\theta \in \Theta} \theta p(\theta y)d\theta$	$\frac{1}{2} = \int_{-\infty}^{\hat{\theta}_{median}} p(\theta y)d\theta = \int_{\hat{\theta}_{median}}^{\infty} p(\theta y)d\theta$
Name	Posterior Var		
Single Para	$C_{\alpha} = [a, b]$ (1 - α)100% Bayesian Credible Intervals (or Simply Credible Intervals)		
Single Para	$Var[\theta y] = \int_{\theta \in \Theta} (\theta - E[\theta y])^2 p(\theta y)d\theta$	$P(a \leq \theta \leq b y) = \int_a^b p(\theta y)d\theta = 1 - \alpha$ ps: $P(\theta < a y) = P(\theta > b y) = \alpha/2$	

Bayesian Decision Theory: 目的: ¹Estimating parameter θ ²Test Hypothesis about θ ³Find CI ps: 后两种有很多种方法来判断

1. Choose a Loss Function $Q(\hat{\theta}, \theta)$ (损失函数, 衡量估计值与真实值的差距; 贝叶斯决策的目的是最小化损失函数的期望)

2. Minimises Posterior Expacted Loss Function: $\hat{\theta}_Q(y) = \arg \min_{\hat{\theta}} E[Q(\hat{\theta}, \theta)|y] = \arg \min_{\hat{\theta}} \int_{\theta \in \Theta} Q(\hat{\theta}, \theta)p(\theta|y)d\theta$

3.2 Exact Bayesian Inference with Conjugate priors

Conjugate Prior: For $\theta, p(\theta)$ is Conjugate Prior for $p(y|\theta) = f(y|\theta)$ iff 对 $p(\theta)$ 用 Bayes' Rule 得到的 $p(y|\theta)$ 都在同一类型的分布

Conjugate Family: The collection of all Conjugate Prior.

Choice of Prior: Observed data number n : ¹ n small $\Rightarrow$ Posterior shows a great dependency of the choice of prior.

² n middle $\Rightarrow$ posterior less affected by choice of prior. ³ n large $\Rightarrow$ choice of prior has limited effect on posterior.(Likelihood has a strong impact)

Comparing the frequentist and Bayesian approaches: ¹ $\hat{\theta}_{MMSE} \approx \hat{\theta}_{MAP}$ for most cases, especially n large. ² n large: $\hat{\theta}_{MAP} \approx \hat{\theta}_{MLE}, CI \approx C_{\alpha}$

3.3 Approximate Bayesian Inference

Intractable Posterior: Non-Conjugate model. (not analytic, no closed-form)

Monte Carlo Methods: Any func $g(\theta) || I_g = \int_{\theta \in \Theta} g(\theta)p(\theta|y)d\theta \approx \frac{1}{M} \sum_{i=1}^M g(\theta^{(i)}) \xrightarrow{a.s.} I_g$ $\theta^{(i)} \sim p(\theta|y), i = 1, ..., M$

1. **Importance Sampling (IS):** ¹ Simulating samples from another distribution $q(\theta)$ (Proposal Distribution) Consistent, Biased for finite

² $w^{(i)} = \frac{\pi(\theta^{(i)}; y)}{q(\theta^{(i)})}$, $\pi(\theta; y) = f(y|\theta)p(\theta)$ ³ IS estimator: $\hat{I}_g^M = \sum_{i=1}^M g(\theta^{(i)})\bar{w}^{(i)} \xrightarrow{a.s.} I_g$, Normalized Weight: $\bar{w}^{(i)} = \frac{w^{(i)}}{\sum_{j=1}^M w^{(j)}}$ $\theta^{(i)} \sim q(\theta)$ ps: $\sum_{j=1}^M \bar{w}^{(j)} = 1$

Remark: Choice of $q(\theta)$ is important. $|| \hat{I}_g^M$ depends on: ¹# of M ; ²mismatch between proposal and the integrand of the integral. Good if $q(\theta) \propto g(\theta)p(\theta|y)$, difficult to achieve.:(

2. **Markov Chain Monte Carlo (MCMC):** MCMC 算法通过 Markov Chain 生成样本, 使得样本接近服从后验分布.(计算量大)

Laplace Approximation: (最简单的估算 posterior 的方法) $p(\theta|y) \approx \mathcal{N}(\theta_{MAP}; H^{-1}(\theta_{MAP}))$ 类似 MLE 的 J $H(\theta) = -\frac{d^2 \ln(p(\theta|y))}{d\theta^2}$ (适合 $p(\theta|y)$ 接近正态分布的时适用)

Variational Inference (VI): ¹Choose $q(\theta; \eta)$, η 是 q 的参数. ² 选择衡量 $p(\theta|y), q(\theta; \eta)$ 差异的方法 e.g.KL divergence: $KL(q(\theta; \eta)||p(\theta|y)) = \int_{\theta \in \Theta} q(\theta; \eta) \ln(\frac{q(\theta; \eta)}{p(\theta|y)})d\theta$

³ 优化参数 (Find parameters to minimise the divergence): $\eta^* = \arg \min_{\eta \in A_{\eta}} KL(q(\theta; \eta)||p(\theta|y))$ ⁴ $p(\theta|y) \approx q(\theta; \eta^*)$

Remark: Both Laplace Approximation & Variation Inference approximate $p(\theta|y)$ by a parameter family of distributions. $||$ 拉普拉斯是正态分布家族, VI 是更普遍的分族

4 Linear Model

4.1 Simple Linear Model (With One Predictor)

Model: $Y_i = \mu_i + \varepsilon_i$, $\mu_i = \beta x_i$ $i = 1, 2, ..., n$ $||$ Suppose ¹ ε_i independent ² $E[\varepsilon_i] = 0$ ³ $Var[\varepsilon_i] = \sigma^2$ $||$ Y_i response variable, x predictor variable.

Least Square (LS) Estimation	RSSResidual Sum of Squares : $S(\beta) = \sum (y_i - \mu_i)^2 = \sum (y_i - x_i \beta)^2$	LS Method: $\hat{\beta}_i = \arg \min_{\beta} S(\beta)$ ps: 求偏导 =0 $ \hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2}$
$E[\hat{\beta}], Var[\hat{\beta}], Var[\hat{\beta}]$	$E[\hat{\beta}] = \beta_{unbiased}$ $ Var[\hat{\beta}] = \sigma_{\hat{\beta}}^2 = (\sum x_i^2)^{-1} \sigma^2$ (σ^2 known)	$Var[\hat{\beta}] = \sigma^2_{\hat{\beta}} = (\sum x_i^2)^{-1} \cdot \sigma^2$ $ \sigma^2 = \frac{1}{n-1} \sum (y - x_i \hat{\beta})^2$
H_0, H_1 Test $ $ CI	$T = (\hat{\beta} - \beta_0) / Var(\hat{\beta}) \sim \mathcal{N}(0, 1)$ (σ^2 known)	$T = (\hat{\beta} - \beta_0) / Var(\hat{\beta}) \sim t_{n-1}$ (σ^2 unknown)
Other	Fitted Values: $\hat{\mu}_i = x_i \hat{\beta}$ $ $ Residuals: $\hat{\varepsilon}_i = y_i - \hat{\mu}_i$	For CI / H_0/H_1 Test: Assume $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$

4.2 General Linear Model

Linear Model: $Y_i = \mu_i + \varepsilon_i$, $i = 1, 2, ..., n$ $||$ Relationship between Parameters: $\underline{\beta} \in \mathbb{R}^p = (\beta_0, ..., \beta_{p-1})$ and response are linear.

· **Matrix Expression:** $\underline{y} = \underline{X}\underline{\beta} + \underline{\varepsilon}$ $||$ Response Vector: $\underline{y}$ | Design / Model Matrix: $\underline{X}$ | Parameters / Coefficients: $\underline{\beta}$ | Additive Noise Vector: $\underline{\varepsilon}$

· **Some Defs:** Parameter Estimates: $\hat{\underline{\beta}}$ | Estimated / Fitted Value Mean Vector: $\hat{\underline{\mu}} = \underline{X}\hat{\underline{\beta}}$ | Estimated / Fitted Residual Vector: $\hat{\underline{\varepsilon}} = \underline{y} - \hat{\underline{\mu}}$

Lamma $|E, Var$ in Matrix: If r.v. $\underline{v}$, $\underline{A}, \underline{b}$ constant $\Rightarrow$ ¹ $E[\underline{A}\underline{v} + \underline{b}] = \underline{A}E[\underline{v}] + \underline{b}$ ² $Var[\underline{A}\underline{v} + \underline{b}] = \underline{A}Var[\underline{v}]\underline{A}^T$

Metric Predictor: 设计矩阵用连续观测值. $||$ **Factor Predictor:** 离散数据用指示函数 (0/1) 填满, 每个"小组" 对应一个指示函数.

4.3 Linear Model Theory

Least Square (LS) Approach:

4.4 Check || Intervals || Testing

4.5 Assumptions || Diagnostics

Graphical Assessment

- 1. Residuals vs Fitted (Check linear assumption): $(\hat{y}_i, e_i)$, $e_i = y_i - \hat{y}_i$, $\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$
在 0 附近,越靠 0 越好。 || 离 0 远: something missing in the model. || 有趋势: Not Constant Var. || ps: 点在整张图均匀分布 $\rightarrow$ constant variance
- 2. Scale-Location (Check constant variance assumption): $(\hat{y}_i, \sqrt{|e_i^*|})$, $e_i^* = \frac{e_i}{s_e}$ 越靠近 0.82 越好.(around a constant value)
- 3. Normal QQ-Plot (Check Normal Assumption): (z_i, e_i^*) , $z_i = \Phi^{-1}(\frac{i-0.5}{n})$

5 Distribution Table

Distribution	PDF	Mean	Var	Other
Bernoulli $X \sim Bern(p)$	$p^x(1-p)^{1-x}, x \in \{0, 1\}$	p	$p(1-p)$	If X_i are i.i.d. $\sum X_i \sim Bin(n, p)$
Binomial $X \sim Bin(n, p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	$X, Y \sim Bin(n, p), Bin(m, p)$ $X + Y \sim Bin(n + m, p)$
For $X \sim Bin(n, p), n \rightarrow \infty, p \rightarrow 0, np \rightarrow \lambda$, then $X \sim Poisson(\lambda)$		For $X \sim Bin(n, p), n \rightarrow \infty$ and pdf symmetric, $\Rightarrow X \sim N(np, np(1-p))$		
Geometric $X \sim Geom(p)$	$(1-p)^{x-1}p$	$1/p$	$(1-p)/p^2$	$E[X^2] = 2/p^2$
Negative Binomial $X \sim NB(r, p)$	$\binom{r+x-1}{x}(1-p)^r p^x, r \geq 1$	$\frac{rp}{1-p}$	$\frac{rp}{(1-p)^2}$	或者 $p = 1 - q$, 注意!
Poisson $X \sim Po(\lambda)$	$e^{-\lambda} \cdot \frac{\lambda^x}{x!}$	λ	λ	$E[X^2] = \lambda^2 + \lambda$
For $X, Y \sim Po(\lambda_1), Po(\lambda_2)$, then $X + Y \sim Po(\lambda_1 + \lambda_2)$				
Multinomial $X_1 + \dots + X_k = n$ $(X_1, \dots, X_k) \sim MN(n, p_1, \dots, p_k)$	$\frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}, 0 < p_i < 1$	np_i	$np_i(1-p_i)$	
Uniform $X \sim \mathcal{U}([a, b])$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$E[X^2] = \frac{b^2+ab+a^2}{3}$
Normal $X \sim \mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi}\sigma^2} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	$\mathbf{Y} \sim N_k(\mu, \Sigma), \mu$ vector $f_{\mathbf{Y}}(y_1, \dots, y_k) = \frac{e^{-\frac{1}{2}(\mathbf{y}-\mu)^T \Sigma^{-1}(\mathbf{y}-\mu)}}{\sqrt{(2\pi)^k det(\Sigma)}}$
For $X, Y \sim N(\mu_1, \sigma_1^2), Po(\mu_2, \sigma_2^2)$, then $aX \pm Y \pm c \sim Po(a\mu_1 \pm b\mu_2 \pm c, a^2\sigma_1^2 + b^2\sigma_2^2)$				
Exponential $X \sim Exp(\lambda)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$E[X^n] = \frac{n!}{\lambda^n}$
Beta $X \sim Beta(\alpha, \beta), \alpha, \beta > 0$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, x \in [0, 1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
When $Beta(\alpha, \beta)$ has maximum probability, $x = (\alpha - 1)/(\alpha + \beta - 2)$				
Gamma $X \sim \Gamma(\alpha, \beta), \alpha, \beta > 0$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x \geq 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\Gamma(\alpha) = (\alpha - 1)!$ $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$
Student's t $X \sim t_v, v > 0$	$\frac{1}{\sqrt{v}B(\frac{v}{2}, \frac{1}{2})} (1 + \frac{x^2}{v})^{-\frac{v+1}{2}}$	0	$\frac{v}{v-2}$	If $Z \sim N(0, 1), Y \sim \chi_n^2$ $\frac{Z}{\sqrt{Y/n}} \sim t_n$
PDF Symmetric at zero and has more mass in the tails (compare to the standard normal distribution)				
When $n \rightarrow \infty$ it will converges to $N(0, 1)$. Cauchy Distribution: $X \sim t_1$ and its mean and var are undefined.				
Chi-Squared $X \sim \chi_v^2, v > 0$	$\frac{1}{2^{\frac{v}{2}} \Gamma(\frac{v}{2})} x^{\frac{v}{2}-1} e^{-\frac{x}{2}}, x \geq 0$	v	$2v$	-
If $X_i \sim N(0, 1)$, then $\sum X_i^2 \sim \chi_n^2$ If $X \sim \chi_n^2, Y \sim \chi_m^2$, then $X + Y \sim \chi_{n+m}^2$				
F $X \sim F(\lambda, v), v > 0$ ps: $f_{a,b;\alpha} = 1/f_{a,b;1-\alpha}$	$\frac{\lambda^{\frac{\lambda}{2}} v^{\frac{v}{2}}}{B(\frac{\lambda}{2}, \frac{v}{2})} x^{\frac{\lambda}{2}-1} (\lambda x + v)^{-\frac{\lambda+v}{2}}, x \geq 0$	$\frac{v}{v-2}$	$\frac{2v^2(\lambda+v-2)}{\lambda(v-2)^2(v-4)}, v > 4$	$Y_1, Y_2 \sim \chi_{n_1}^2, \chi_{n_2}^2, X = \frac{Y_1/n_1}{Y_2/n_2}$ $X \sim F_{n_1, n_2}, X^{-1} \sim F_{n_2, n_1}$

6 Review

Law of Total Probability: $\mathbb{P}(A) = \sum_n \mathbb{P}(A|B_n)\mathbb{P}(B_n)$, $\cup_n B_n = \Omega$, B_n disjoint

Type	Type I error	Type II error	-	H_0 is T	H_0 is F
Def	$\mathbb{P}(Rej H_0 H_0 \text{ is } T)$	$\mathbb{P}(Fail Rej H_0 H_0 \text{ is } F)$	Fail to Rej H_0	Right	Type II (β) 与 size 有关
			Rej H_0	Type I (α) 与 size 无关	Right (Power)

Significance Level: $\alpha = \mathbb{P}(Rej H_0 | H_0 \text{ is } T)$

P-value: 1.One-side: $p = \mathbb{P}(T \geq t; T \sim H_0)$ 2.Two-sides: $p = 2 \min\{\mathbb{P}(T \geq t; T \sim H_0), \mathbb{P}(T \leq -t; T \sim H_0)\}$

Power of Test: $\mathbb{P}(Rej H_0 | H_0 \text{ is } F) = 1 - \beta = \mathbb{P}(Z \in C | \mu = \mu^*) \cdot 1.$ Sample size $\uparrow$, Power $\uparrow$; 2. $\sigma^2 \downarrow$, Power $\uparrow$; 3. Effect size $|\theta_0 - \theta^*| \uparrow$, Power $\uparrow$; 4. $\alpha \uparrow$, Power $\uparrow$.

Def of Confidence Interval (CI): 100(1 - α)% Confidence Interval for θ is $[l(X), u(X)]$, $\mathbb{P}(l(X) < \theta < u(X)) = 1 - \alpha$.

(Convergence of r.v.) Name	Definition	written as	Example
Convergence in Probability	If $\lim F_{X_n}(x) = F_X(x)$ for all $x \in \mathbb{R}$.	$X_n \xrightarrow{d} X$ or $X_n \xrightarrow{d} F_X$	CLT 最弱; 见下面

Convergence in Distribution	If $\forall \epsilon > 0, \lim \mathbb{P}(X_n - X > \epsilon) = 0.$	$X_n \xrightarrow{p} X$	WLLN 中等:	If $X_i \overset{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu.$ $\Rightarrow \bar{X}_n \xrightarrow{p} \mu$
Almost Sure Convergence	If $\mathbb{P}(\lim X_n = X) = 1$	$X_n \xrightarrow{a.s.} X$	SLLN 最强:	If $X_i \overset{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu,$ $Var[X_i] < \infty. \Rightarrow \bar{X}_n \xrightarrow{a.s.} \mu$

Central Limit Theorem (CLT): If $X_i \overset{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$ and $Var[X_i] < \infty.$ $\Rightarrow \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \overset{d}{\sim} N(0, 1)$ 样本均值的分布会趋近于正态分布

7 Writting Format

Distribution: 产生新 distribution 的一种方法: 通过条件概率来. $\mathbb{P}(A|B) = \mathbb{P}[(A \cap B)/B]$

p-value: p-value measures the plausibility that the data were generated under H_0 . **Low p-value:** low p-value suggests H_0 is incompatible with the data and hence likely to be false.

CI: The set of values (e.g. θ) are not rejected at significance level α under H_0 . If data are generated under H_0 , there is a probability $(1 - \alpha)$ for the true θ_0 to be within the $(1 - \alpha)100\%$ CI.

Hypothesis Testing: **Steps:** ¹ State H_0, H_1 ; ² Select Test Statistic T ; ³ distribution of T under H_0 ; ⁴ observed test statistic; ⁵ critical value / region, p-value, or CI; ⁶ conclusion. **Reject H_0 :** We would reject H_0 in favour of the H_1 . (So that ...) **Fail to reject H_0 :** We fail to reject H_0 . Hence, there is strong evidence against the null hypothesis.

MLE: Using (Compare) Newton's Method = ($\neq$) Scoring Method: Because the second derivative of likelihood function doesn't (does) include the r.v.

MLE Inference Parameter $|\mathbb{P}(\theta < ?) = \alpha$ is false: We cannot make such probability statement. $\{\theta\}$ is a fixed value (but unknown).

Geometric Comparison of $T_{Score}, T_{Wald}, T_{LR}$: **1. T_{Score}** (Slope Comparison): Represents the slope of the likelihood function at θ_0 ; A steeper slope indicates that the likelihood function is changing more rapidly at θ_0 , which implies a greater difference between $\hat{\theta}$ and θ_0 . **2. T_{Wald}** (Width Comparison): Compares the difference between $\hat{\theta}$ and θ_0 . **3. T_{LR}** (Height Comparison): Compares the maximum likelihood value $L(\hat{\theta})$ and the likelihood value at $\theta_0, L(\theta_0)$. A larger height difference indicates a greater discrepancy between $\hat{\theta}$ and θ_0 **$T_{Score}, T_{Wald}, T_{LR}$ Other** 部分公式可以近似的理由: This is due to $\hat{\theta}$ being a consistent estiamtor of θ_0 , and $J(\theta_0)$ being a consistent estimator of $I(\theta_0)$

Normal Linear Regression Assumptions: Assumptions are that the observations: 1. are from a normal distribution; 2. are independent; 3. have a common variance; 4. Linearity of the expectation variable.

Linear Model: LS Estimation: $S(\beta)$ small: μ_i relatively close to the y_i ; Large: μ_i far from their corresponding y_i . || **Meaning of ε :** The random component of the model is intended to capture the fact that if we gathered a replicate set of data, the β_i (coefficient) would not change, but the Y_i would, as a result of different measurement errors. || **Why Use Least Square to estimate $\underline{\beta}$:** Most of time, Linear Model can be a system of n equations with p parameters with $n > p$. In this case, the system is overdetermined and there is no exact solution. The least square method is used to find the best solution.